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Circular Value Propositions for Second-Life Battery Energy Storage Solutions: An Explorative Study of Charging Infrastructure

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Abstract. The large-scale electrification of transport is accelerating globally, resulting in a growing influx of end-of-first-life batteries from electric vehicles (EVs). Repurposing these batteries into second-life battery energy storage systems (SL-BESS) offers a promising pathway to extend battery lifecycles, reduce environmental impacts, and support circular value creation. Despite significant technological progress and numerous pilot projects, the adoption of SL-BESS in scalable business practice remains limited. A key challenge lies in defining viable and context-specific value propositions that can guide actors in designing circular business models. This paper presents an explorative qualitative study examining how circular value propositions for SL-BESS can be defined and contextualized within EV charging infrastructure. Empirical data was gathered through semi-structured interviews and co-creative workshops involving multiple actors in the Swedish EV ecosystem. The analysis resulted in three business scenarios: resilient energy hubs within municipalities, mobility-integrated energy for flexible and bidirectional bus charging, and circular Battery-as-a-Service solutions for industrial and mining applications. The findings indicate that while SL-BESS technologies have reached a relatively high level of maturity, business model development lags due to regulatory uncertainty, fragmented market structures, and unclear revenue and ownership arrangements. By abstracting across scenarios, the study identifies recurring patterns in how second-life batteries can function as enabling infrastructure for flexible operations, shared resource use, and industrial–societal symbiosis. The paper contributes to the circular business model literature by presenting scenario-based circular value propositions, cross-scenario comparison criteria, and strategic patterns that can support actors in positioning themselves within emerging markets for circular energy storage and charging infrastructure.

Keywords: Second-life batteries, Battery circularity, Circular business models, Value proposition design.

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1. Introduction

The transport sector is a critical component of the European Union's ambition to achieve climate neutrality by 2050 [1]. Electrification of vehicles, driven by policy mandates and technological innovation, has accelerated rapidly in recent years. In Sweden, transportation accounts for nearly one-third of total greenhouse gas emissions [2]. Achieving the national target of carbon neutrality by 2045 therefore requires a transformative shift toward electrified mobility, with an estimated 2.5 million electric vehicles (EVs) expected on Swedish roads by 2035 [3]. At the global level, the number of EVs is projected to exceed 145 million by 2030 [4].

While increasing EV adoption is a key step toward decarbonization, it also introduces new challenges related to battery lifecycle management and charging infrastructure capacity. EV batteries typically lose 20–30% of their capacity after several years of use, at which point they are no longer suitable for automotive applications but still retain substantial energy storage capability [5]. By 2030, an estimated 100–200 GWh of end-of-first-life EV batteries will become available annually, a volume nearly equivalent to current global battery production [6]. Managing and reusing this growing stock of retired batteries is therefore central to achieving a sustainable and circular energy transition. At the same time, rising EV penetration places increasing pressure on already constrained local electricity grids. Swedish grid operators report growing difficulties in meeting the power capacity demands of urban charging infrastructure, particularly in cities such as Stockholm, Gothenburg, and Luleå [7]. Developing supplementary charging solutions that enhance flexibility and grid stability is therefore essential. Second-life batteries, repurposed as stationary battery energy storage systems (BESS), offer a promising means to balance supply and demand, integrate renewable energy, and provide resilience services for both public and industrial users [8].

Battery circularity refers to extending battery lifecycles through reuse, remanufacturing, repurposing, and eventual recycling [9]. For EV batteries, second-life applications are increasingly viewed as a critical intermediate step before recycling, as they can enable grid stabilization, improve renewable energy utilization, and reduce carbon and resource intensity associated with new battery production. Harper et al. [8] estimate that up to 85% of EV batteries are suitable for second-life use, with potential additional lifespans of 5–15 years depending on application context.

From the perspective of industrial and societal symbiosis [10], second-life battery systems enable new forms of resource sharing between municipalities, transport operators, and industrial production systems. Energy storage capacity, charging infrastructure, and operational flexibility can be jointly utilized across organizational boundaries, transforming energy from a fixed external constraint into a configurable element of infrastructure and production system design. As the scaling of EV charging infrastructure must progress in tandem with vehicle adoption, these characteristics are increasingly important. As of early 2023, Sweden hosted over 19,000 public charging points serving approximately 445,000 passenger EVs, 13,700 light trucks, and 940 electric buses [11]. Delays in grid expansion risk slowing both climate targets and industrial electrification efforts. In this context, battery energy storage, particularly second-life solutions, can function as a transitional technology that supports grid reinforcement, charging flexibility, and more resilient operations [12].

Despite these opportunities, existing research has primarily focused on the technical feasibility of second-life battery applications, including safety, degradation behavior, and lifecycle impacts [13]. Comparatively less attention has been given to how second-life battery systems can be translated into viable business models and value propositions, particularly within the EV

charging ecosystem where multiple actors—municipalities, transport operators, utilities, and manufacturers—interact [14]. The absence of clear, actor-specific value propositions remains a key barrier to commercialization beyond pilot projects [15].

Against this background, this paper explores how circular value propositions for second-life battery energy storage systems can be defined and contextualized for different actors within the charging infrastructure ecosystem. The study is guided by the following research question: *How can circular value propositions for second-life battery energy storage systems be defined and contextualized for different actors within the charging infrastructure ecosystem?* Adopting an explorative approach, the paper does not aim to predict market outcomes but instead identifies business scenarios and value creation logics that can support emerging open markets for second-life batteries and inform strategic positioning among public and industrial stakeholders.

2. Theoretical background

2.1 Second-life batteries

The circular economy has become a central paradigm for addressing environmental and resource challenges in industrial systems. Within the battery sector, circularity emphasizes maximizing material value across multiple lifecycles while minimizing waste and environmental impact. This policy is reinforced by the European Union's Battery Regulation (EU) 2023/1542, which mandates traceability, reuse, and recycling, signaling a systemic shift toward circular battery management [16]. Second-life battery energy storage systems (SL-BESS) can provide services such as peak shaving, load balancing, renewable energy integration, and backup power across residential, commercial, and industrial contexts [5][12]. Several studies also demonstrate environmental benefits: reuse can reduce lifecycle greenhouse gas emissions by up to 30% compared to direct recycling and delay the extraction of critical raw materials such as lithium, cobalt, and nickel [6] [17].

Over the past decade, the focus of literature has shifted markedly. Early research concentrated on technical feasibility, addressing degradation uncertainty, safety, and reconfiguration costs. More recent contributions emphasize that technical maturity alone is insufficient for large-scale adoption. Instead, challenges related to ecosystem coordination, responsibility allocation, standardized qualification, and viable revenue logic have become central [18]. Supply-chain and governance studies further highlight the need for industrialized reverse logistics and transparent accountability structures to move beyond pilot projects [19].

Despite this progress, the literature largely treats second-life batteries either from a technical optimization perspective or through firm-centric business models [12]. Less attention has been given to how second-life battery value is co-created across multiple actors [9][14]—such as municipalities, transport operators, utilities, and industrial firms—particularly in charging infrastructure contexts where public and private interests intersect. This gap is especially relevant from the perspective of industrial and societal symbiosis [10], where energy storage capacity, infrastructure, and operational flexibility are shared across organizational boundaries.

2.2 Business models in circular systems

A central challenge in realizing circular economic potential lies not in technology, but in business models. Geissdoerfer et al. [20] define circular business models as those that create, deliver, and capture value while closing material loops and decoupling growth from resource consumption. In battery systems, this implies shifts toward service-based offerings, new ownership arrangements, and collaborative ecosystems rather than linear product sales [9] [15]. Value proposition design

plays a critical role in this transition. The Value Proposition Canvas (VPC) developed by Osterwalder and Pigneur [21] provides a structured method for aligning customer jobs, pains, and gains with products and services. In circular contexts, however, value propositions must extend beyond single customers to include environmental and social value dimensions [22]. Chirumalla et al. [9] further identify three circular business model archetypes for batteries—extending, sharing, and looping —and emphasize that their successful operationalization depends on diverse forms of collaboration among actors. Nevertheless, circular business models remain characterized by uncertainty [24]. Profitability depends on volatile material markets, evolving regulation, and complex reverse logistics. This uncertainty is particularly pronounced for second-life batteries, where ownership, liability, and revenue distribution remain unclear. Service-based models such as Battery-as-a-Service (BaaS) have therefore emerged as mechanisms to redistribute risk and support lifecycle responsibility [25].

2.3 Energy storage and charging infrastructure

Energy storage systems are critical enablers of the energy transition in both stationary and mobile contexts. The Vehicle-to-Grid (V2G) concept introduced by Kempton and Tomić [26] illustrates how EVs can contribute to grid balancing by providing bidirectional energy flows. Subsequent research highlights the role of integrated storage solutions in managing peak loads, supporting frequency regulation, and reducing dependence on grid upgrades [27] [28]. Within charging infrastructure, the integration of BESS and V2G technologies enables more flexible and resilient operations. These systems increasingly function as shared assets within urban and industrial energy systems, requiring coordination among municipalities, transport operators, and energy providers [29]. Such coordination introduces governance challenges related to data sharing, tariff design, and investment responsibility, reinforcing the need for business models that account for multi-actor value creation [18] [30].

2.4 Extending the VPC for multi-actor circular systems

The VPC aligns a customer segment with a value proposition by mapping customer jobs, pains, and gains to products and services, pain relievers, and gain creators [21]. Traditionally, the VPC is applied to single-firm, single-customer contexts, where value creation and capture are largely internal to one organization. Recent studies demonstrate that the VPC can be adapted beyond this original scope. Pokorná et al. [31] apply the VPC to consumer behavior in local food markets, while Meng et al. [32] reinterpret policy measures as value propositions in mobility systems. Islam and Iyer-Raniga [33] further extend the VPC to incorporate environmental and social stakeholders, reflecting triple-bottom-line value creation in circular business models.

However, even these extensions tend to conceptualize value propositions at the level of individual actors. What remains underexplored is how the VPC can be used to articulate multi-actor circular value propositions [9][22][23][33], where value is jointly created, delivered, and captured across interconnected public and private actors. This limitation is particularly evident in second-life battery systems, where benefits such as grid stability, resilience, and emission reduction accrue unevenly across stakeholders.

In this study, the VPC is used not only as an analytical tool but as a co-creative tool to compare and synthesize actor-specific value logics. By applying the VPC across municipalities, transport operators, energy providers, and industrial firms, the study extends the canvas from a firm-centric design tool to a mechanism for exploring ecosystem-level circular value propositions. This approach addresses a gap in the literature by showing how classic VPC logic can be adapted to

support multi-actor coordination and industrial–societal symbiosis [10] in emerging circular energy systems.

3. Research Methodology

3.1 Research approach and setting

This study adopts a qualitative, exploratory research design [34] to investigate how value propositions for SL-BESS can be conceptualized and contextualized. An exploratory approach is appropriate when a phenomenon is emergent, lacks clearly established theoretical boundaries, or remains underexplored in practice [34]. Given the novelty of SL-BESS business models within charging infrastructure, the research follows a constructivist paradigm, assuming that knowledge about circular value propositions is co-produced through interaction between researchers and practitioners rather than derived through hypothesis testing. The analytical focus is therefore on sensemaking—how different actors interpret challenges, opportunities, and value creation logics within a complex, multi-actor ecosystem.

Empirically, the study was conducted within the research project SmartCharg (Smart and Circular Battery Charging Solutions). The project consortium comprised seven organizational actors representing key roles in the Swedish charging and electromobility ecosystem. These included two municipalities responsible for public infrastructure and energy planning (Companies A and B), one energy service provider specializing in grid balancing and distributed energy management (Company C), three public transport operators managing electric bus fleets and depot charging systems (Companies D, E, and F), and one vehicle original equipment manufacturer (OEM) operating in the mining sector with an emerging interest in circular battery solutions for off-grid energy use (Company G). This configuration reflects the interconnected public and private actors involved in charging infrastructure development and second-life battery ecosystem. A purposeful sampling strategy was applied [35] to ensure relevance to the research aim while capturing diversity in organizational roles, regulatory environments, and operational energy needs. This setting enabled the study to examine circular value propositions not at the level of individual firms, but across actor constellations that jointly shape charging infrastructure and energy storage practices.

3.2 Data collection

Given the exploratory aim of the study, a combination of semi-structured interviews and co-creative workshops was used to capture early-stage insights into emerging SL-BESS business models and actor-specific value propositions. The empirical design consisted of three complementary phases that together enabled triangulation between individual perspectives, structured value proposition mapping, and collective scenario development.

In the first phase, ten semi-structured interviews were conducted with representatives from the seven partner organizations (Companies A–G), including municipalities, public transport operators, an energy service provider, and an industrial OEM (see Table 1). The interviews aimed at understanding current battery-related operations and identifying potential application areas for SL-BESS. The interview guide covered themes related to current battery operations and charging, existing experiences with BESS, perceived potential of second-life batteries within an ecosystem, two prioritized SL-BESS application areas, SWOT reflections for these applications, preconditions for development of SL-BESS, and relevant regulatory considerations. All interviews were audio-recorded and transcribed verbatim.

The second phase focused on structured value proposition articulation using the VPC [21]. Partner organizations were invited to prepare one or two preliminary VPCs describing SL-BESS applications relevant to their operational context and maturity. To support consistency, the research team provided a short introduction to the VPC (including written and video material), a template, and guiding prompts. In total, six company-specific VPCs were submitted: one from a municipality (Company A), four from the three transport operators (Companies D–F), and one from the industrial OEM (Company G). These canvases were then presented and discussed in a co-creative workshop lasting 5 hours and 30 minutes, attended by seven participants representing the seven organizations (Table 1). Each organization presented its VPC (approximately 10 minutes) followed by a short discussion. After the presentations, participants worked in two mixed groups to jointly develop an additional VPC and a corresponding business scenario, including key preconditions for implementation. This resulted in a total dataset of eight VPCs: six company-specific canvases and two joint canvases developed during the workshop.

In the third phase, the research team synthesized the material into three preliminary scenario-level value proposition configurations and conducted a second co-creative workshop to validate and refine them. This workshop lasted 5 hours and 30 minutes and involved six participants from five organizations (three bus companies, one municipality, and one energy service provider; see Table 1). The workshop theme, strategic business scenarios for future energy storage, supported a deeper discussion of scenario assumptions and enabled stakeholder mapping, as well as the identification of key challenges and enablers. Participants again worked in two mixed groups, supported by the same facilitation structure as in the previous workshop. Guiding questions addressed actor roles (customers, partners, public agencies), actor needs and pain points, required infrastructure (fixed, mobile, and digital), contribution to sustainability or resilience goals, and success factors for adoption. The workshop resulted in stakeholder mappings for two scenarios, including customer needs, perceived challenges, and enabling conditions. All workshop activities were audio-recorded and documented by the research team for subsequent analysis.

Table 1. Detail information about the data collection methods, involved companies, and informants

Type companies	No of participants	Roles	Duration (minutes)
Interviews			
Company A: Municipality	2	Municipality energy strategist, Process leader	1 hr 57 mins, 1 hr 15 mins
Company B: Municipality	1	Unit manager for vehicle services	1 hr 50 mins
Company C: Energy service provider	2	Business advisor, Product manager EV charging	1 hr 30 mins, 1 hr 9 mins
Company D: Buss operator	1	Technology and Fuel strategist	1 hr 50 mins
Company E: Buss operator	2	Head of Innovation, Business manager	1 hr 12 mins, 59 mins

Company F: Buss operator	1	Head of sustainability and innovation	1 hr 33 mins
Company G: OEM	1	Project manager electrification solutions	1 hr 30 mins
		Workshops	
Workshop 1: Companies A, B, C, D, E, F, G	7	Municipality energy strategist, Unit manager for vehicle services, Business advisor, Technology and Fuel strategist, Head of Innovation, Head of sustainability and innovation, Project manager Electrification solutions	5 hr 30 mins
Workshop 2: Companies A, C, D, E,	6	Municipality energy strategist, Sustainability strategist, Product manager EV charging, Technology and Fuel strategist, Head of Innovation, Head of sustainability and innovation.	5 hr 30 mins

3.3 Data analysis

The analysis followed a multi-step process that combined thematic coding with structured synthesis using the VPC dimensions. The goal was to link the empirical material (interviews, VPCs, and workshop discussions) to concepts from circular business model research [9] [15] [22] and to compare value propositions across actor types. Using interviews and workshops together also supported triangulation between individual perspectives and joint scenario work.

First, the interview and workshop transcripts were coded to identify recurring themes related to SL-BESS application areas, VPC dimensions, and criteria used to compare scenarios. In parallel, all eight VPC templates (six company-specific and two group-based) were compiled and then transferred into an Excel matrix. The first column listed the actors (Companies A–G) and the two workshop groups, and the subsequent columns captured the main VPC dimensions: customer segments, customer jobs, pains, gains, products and services, pain relievers, gain creators, and the overall value proposition. This format enabled systematic cross-case comparison. The research team reviewed the matrix column by column and used color coding to mark similar ideas across actors. Based on patterns in the six company-specific VPCs, three preliminary scenario clusters were identified. The two group-based VPCs were analyzed in the same way and added as two additional scenario clusters. This resulted in five preliminary scenarios that reflected distinct SL-BESS opportunities across municipal, transport, and industrial contexts.

Second, the five preliminary scenarios were discussed in several meetings among four researchers. These discussions focused on overlaps, distinguishing features, and actor roles (e.g., who could lead and who could complement) as well as key dependencies and conditions for implementation. Through this consolidation, the five scenarios were refined into three core business scenarios: one centered on municipalities, one on transport operators, and one on industrial and mining applications. These three scenarios were then presented and validated in follow-up workshop 2, where partner companies provided feedback, clarified assumptions, and supported stakeholder mapping of actor roles. Overall, this process combined thematic coding,

structured VPC comparison, and researcher and partner validation to derive the final set of three business scenarios.

4. Results: Three Business Scenarios

The empirical analysis resulted in three business scenarios illustrating how different actors can operate SL-BESS within the evolving electric mobility and charging infrastructure ecosystem. Each scenario corresponds to a distinct stakeholder domain—municipalities, transport operators, and industrial companies—and highlights how value is created, delivered, and captured under different ownership, operational, and regulatory conditions. Rather than prescriptive models, the scenarios are presented as strategic narratives that reflect emerging practices and potential pathways for commercialization.

4.1 Scenario 1: Resilient energy hubs within municipalities

Municipalities in Sweden (Companies A and B) face increasing challenges related to energy reliability, cost stability, and climate resilience. As public infrastructure such as schools, hospitals, and elderly care facilities becomes increasingly electrified, demand for flexible and stable electricity supply grows. These challenges are intensified by local grid constraints, particularly in smaller urban centers, creating a need for decentralized energy storage solutions.

Second-life BESS offers municipalities a scalable means to store locally generated renewable energy, manage peak loads, and ensure continuity of critical services during outages. These solutions align with public procurement objectives under Sweden's national climate strategy, which emphasizes circularity and resource efficiency. In this scenario, municipalities act as local energy integrators, supported by private technology providers that deliver modular, service-oriented SL-BESS solutions. The core value proposition centers on affordable energy stability and resilience, combined with visible leadership in sustainable innovation. Ownership is shared through public–private partnership arrangements, where municipalities collaborate with private firms that provide system integration, operation, and maintenance. The private company assumes the role of energy service integrator and offers Energy-as-a-Service contracts with long-term monitoring and performance guarantees. Revenue streams include service subscriptions, municipal energy-efficiency grants, and participation in grid-balancing markets such as frequency containment services.

The scenario is sensitive to external factors including energy price volatility, public procurement regulations, and rules governing municipal participation in energy markets. Technological developments in battery health diagnostics and recycling also influence long-term economic viability. Overall, this scenario illustrates how municipalities can move beyond passive energy consumption toward active participation in local energy systems, leveraging circular technologies to strengthen resilience and support sustainability transitions.

4.2 Scenario 2: Mobility-integrated energy for flexible and bidirectional bus charging

Public transport operators in Sweden (Companies D, E, and F) are rapidly electrifying their bus fleets to meet emission reduction targets. While electrification reduces operational emissions, it also places significant demands on organizations in terms of planning, investments, and internal resources. Charging depots and operational routes experience high and uneven power demand, creating cost pressures, grid capacity constraints, and operational risks during the transition phase.

From the operators' perspective, this scenario unfolds over two time horizons. In the short term (3–5 years), the primary role of battery energy storage is as an enabler of electrification rather than as a market-facing flexibility asset. Small-scale and mobile BESS are deployed to support decentralized charging along operational routes and at depots with limited grid capacity. In this phase, storage functions mainly as an operational buffer, reducing peak loads, improving charging reliability, and allowing fleet electrification to proceed despite local grid constraints. The focus is on operational continuity and risk reduction rather than energy market participation.

In the longer term (5–10 years), as fleet electrification stabilizes and organizational capacity increases, the scenario evolves toward a more integrated energy system with V2G functionality. Electric buses, combined with stationary SL-BESS at depots, are then treated as mobile and fixed energy assets that interact dynamically with the electricity system. Vehicles can charge during low-price or low-demand periods and discharge during peak demand, while stationary SL-BESS buffers local loads and supports renewable energy integration. These functions are enabled through collaboration with technology platform providers that deliver digital energy management, forecasting, and control capabilities.

Ownership follows a hybrid structure, where transport operators retain ownership of vehicles, while charging and energy optimization capabilities are provided through partnerships with technology companies and, where relevant, energy service providers. Revenue opportunities emerge gradually, starting with operational cost savings from optimized charging and later expanding to grid service payments, flexibility markets, and energy resale. Value is distributed through revenue-sharing arrangements among bus operators, technology partners, and local utilities.

The long-term success of this scenario depends on regulatory approval for bidirectional energy trading, access to dynamic electricity pricing, and the continued development of interoperable standards such as ISO 15118. Market incentives for flexibility services are also critical. Overall, the scenario illustrates a staged transition in which mobility operators first use BESS to enable electrification under constrained conditions, and later evolve into active participants in a more flexible and circular energy system.

4.3 Scenario 3: Circular battery-as-a-service (BaaS) for industrial and mining sectors

Industrial and mining companies (Company G) operate in energy-intensive environments, often in remote or grid-constrained locations. These actors face dual pressures to ensure reliable energy supply and comply with increasingly stringent sustainability and reporting requirements.

SL-BESS provides a low-carbon alternative to conventional diesel generators by supplying modular and, where needed, mobile energy storage units. In this scenario, a service provider offers BaaS, covering installation, operation, maintenance, and end-of-life management of second-life batteries. Customers pay for energy or capacity as a service, while the provider retains ownership and lifecycle responsibility. Retaining ownership enables traceability and supports circular economy principles by shifting investment risk from the customer to the provider and emphasizing performance-based consumption. Revenue is generated through pay-per-use contracts, service subscriptions, and value-added data services such as predictive maintenance and carbon tracking, which enhance operational transparency.

The viability of this model depends on regulatory conditions, the availability of certified second-life batteries, advances in diagnostics and recycling technologies, and local energy market conditions. From a production perspective, the BaaS model supports stable and predictable energy supply, particularly in remote production sites. By externalizing energy asset ownership

and maintenance, industrial firms can focus on core production activities while improving operational uptime, flexibility, and compliance with sustainability requirements.

4.4 Cross-scenario comparison and synthesis

Taken together, the three scenarios represent different forms of industrial–societal symbiosis, where surplus battery capacity from electric vehicles is repurposed to support public infrastructure, mobility operations, and industrial sites. Although the actors, ownership structures, and revenue models differ, all scenarios position SL-BESS as enabling infrastructure that links circular resource use with charging infrastructure needs and operational requirements. Table 2 summarizes the key differences and similarities.

Table 2. Cross-scenario comparison matrix

Criterion	Scenario 1: Resilient Energy Hubs within Municipalities	Scenario 2: Mobility-Integrated energy system	Scenario 3: Industrial BaaS
Ownership	Shared (public-private)	Hybrid (fleet + platform)	Provider-owned
Lead/ Complement role	Company leads as system integrator	Company co-led with technology platform providers	Company leads as service operator
Time Horizon	Long-term infrastructure (5-15 years)	Short–medium term (3–5 yrs) enabling electrification; long term (5–10 yrs) V2G integration	Long-term service lifecycle
Deployment Nature	Permanent	Short term: mobile / decentralized BESS; long term: semi-permanent depot-based	Temporary/relocatable
Revenue Model	Energy-as-a-service contracts, municipality energy efficiency grants, grid services	Short term: operational cost savings; long term: grid services, flexibility markets, energy resale	Pay-per-use contracts, service subscriptions, value-added data services (e.g., predictive maintenance, carbon tracking)
External Factors	Energy prices, public procurement, policy mandates	Grid capacity, organizational readiness, electricity market rules, V2G standards	Regulatory compliance, recycling and diagnostics technology
Primary Value Driver	Local energy resilience	Enabling electrification and later grid flexibility	Reliable, sustainable off-grid power

Across the scenarios, **ownership** arrangements vary in a way that reflects risk allocation and responsibility. Municipal Energy Hubs rely on shared ownership in public–private partnerships,

Mobility-Integrated V2G combines fleet ownership with provider-managed platform assets, and Industrial BaaS retains provider ownership to ensure traceability and lifecycle control. In all three cases, the focal company takes a leading role, but with different functions: integrator in the municipal setting, platform provider in the V2G setting, and service operator in the industrial setting.

The scenarios also differ in **time horizon** and **deployment character**. Municipal Energy Hubs are long-term and permanent infrastructure embedded in public facilities. V2G is more operational and flexible, typically based on semi-permanent or mobile configurations linked to depot conditions and fleet operations. Industrial BaaS is based on long-term service contracts, but the technical units are often relocatable to match changing site needs.

Revenue logic follows these differences. Municipal solutions combine Energy-as-a-Service contracts with public grants and grid services. V2G value capture depends on revenue sharing from flexibility markets, price arbitrage, and operational savings from charging optimization. Industrial BaaS relies on pay-per-use and subscription models, often complemented by data-driven services such as predictive maintenance and sustainability reporting.

External dependencies shape feasibility across contexts. Municipal applications are strongly influenced by public procurement rules, policy mandates, and energy price development. V2G depends on access to electricity markets, bidirectional trading permissions, and interoperability standards. Industrial BaaS is particularly sensitive to regulatory compliance demands, the availability of certified second-life battery supply, and advances in diagnostics and recycling.

Despite these differences, three cross-scenario insights stand out. First, all scenarios show a shift from product delivery to service-oriented value creation, where performance and lifecycle management are central. Second, SL-BESS business viability depends on coordinated ecosystems rather than single-firm action, since value creation and risk are distributed across public and private actors. Third, circularity becomes more than an environmental goal: it supports resilience, flexibility, and cost control, and thereby strengthens competitive positioning. Overall, the comparison indicates that SL-BESS should be understood not only as energy assets but as infrastructure that enables more flexible charging and operations under grid constraints.

The cross-scenario synthesis also reveals three recurring strategic patterns for operationalizing circularity in the battery and energy sectors. The first is an **Integrator strategy**, exemplified by the municipal scenario, where firms act as system integrators that combine hardware, software, and maintenance services into modular solutions tailored to local infrastructure and energy needs. The second is a **Platform strategy**, evident in the mobility scenario, in which companies function as energy brokers in collaboration with technology platform providers, managing bidirectional energy flows, monetizing flexibility, and coordinating fleets, grids, and markets through digital platform. The third is an **Operator strategy**, observed in industrial applications, where providers retain asset ownership and deliver turnkey energy services with full lifecycle responsibility, aligning closely with emerging sustainability, traceability, and compliance requirements. Taken together, these strategies illustrate a broader servitization pathway for the battery industry, indicating a structural shift in value creation from manufacturing-centered models toward continuous service provision and lifecycle stewardship.

5. Discussion

5.1 Theoretical contributions

This study contributes to research on battery circularity and circular value creation by showing how value propositions for SL-BESS take shape across multiple actors, not within a single firm [9][14]. The three scenarios illustrate that value is co-created through networks [18] that include municipalities, transport operators, utilities, recyclers, technology providers, and industrial users. This supports and extends earlier circular economy work that emphasizes coordination and shared responsibility in complex systems [9] [22], and it makes these ideas concrete in the context of charging infrastructure and second-life batteries.

A second contribution is the clear shift from product delivery to service-based value creation [15]. Across all scenarios, the value proposition depends less on selling storage hardware and more on delivering outcomes over time, such as resilience, flexibility, and reliable power. Energy-as-a-Service, Vehicle-to-Grid participation, and Battery-as-a-Service show how circular principles are operationalized through contracts where performance, maintenance, and end-of-life responsibility are managed over the lifecycle rather than transferred at sale. This reinforces the view that circularity is not only about material recovery, but also about new ways of delivering and capturing value that support longer use and lower risk for users.

The scenarios also help explain why SL-BESS adoption remains limited despite technical progress. While pilots demonstrate feasibility, the step from pilots to scale is blocked by uncertain rules, unclear ownership and liability, and fragmented revenue mechanisms. The scenarios show that defining actor-specific value propositions is a key step in reducing this uncertainty because it clarifies who benefits, who pays, and what conditions must be in place for implementation.

Beyond the energy domain, the findings also speak to production development and operations. In all three scenarios, SL-BESS acts as enabling infrastructure that buffers energy variability, supports peak management, and improves operational flexibility under grid constraints. This affects how organizations plan operations, manage uptime, and expand capacity, especially in energy-intensive and electrified environments.

Finally, from an open market perspective, the scenarios should be read as value proposition configurations rather than market forecasts. If used batteries increasingly become tradable assets, market formation will depend not only on volumes and prices but also on clear roles, ownership structures, and coordination mechanisms. By describing these elements, the scenarios contribute to market formation by reducing uncertainty for actors considering entry into second-life battery markets and by highlighting practical requirements for transparent and competitive exchange.

5.2 Managerial implications

The study shows that second-life battery energy storage systems SL-BESS are not only sustainability solutions but also practical tools for improving charging and operational performance under grid constraints. For managers, the first actionable step is to treat the value proposition as a design choice that must specify three elements clearly: the target actor and use case (e.g., municipal resilience, depot peak management, off-grid industrial reliability), the operating model (who owns, operates, and maintains the system), and the revenue logic (service fees, flexibility markets, savings, or shared value). Making these elements explicit helps reduce uncertainty and avoids pilot projects that lack a pathway to scale.

For companies developing SL-BESS offerings, the results suggest three concrete capability priorities. First, firms need robust battery qualification and system design routines, including diagnostics, module selection, and safety validation, because trust and performance guarantees are central to service-based models. Second, firms should build operational capabilities for monitoring, maintenance, and performance reporting, since most customers value uptime, predictable costs, and reduce operational risk more than the hardware itself. Third, firms should develop contracting and partnership templates that clarify liability, data access, and end-of-life responsibility, because unclear responsibility is a recurring barrier to implementation across scenarios.

• For public authorities, the findings point to specific changes in procurement and project setup. Municipalities can increase adoption by shifting from equipment purchase to performance-based service contracts where suppliers are evaluated on outcomes such as resilience, peak reduction, and lifecycle management. Municipal actors can also accelerate implementation by pre-defining acceptable partnership structures and responsibilities (e.g., who participates in grid services, who receives revenues, and who carries operational risk). In addition, municipalities can use pilot projects more strategically by requiring scalability conditions from the start, such as standardized testing procedures, interoperability requirements, and measurable performance indicators.

• For transport operators, the scenarios highlight the operational value of combining depot charging, stationary storage, and bidirectional flexibility, but only if market access and coordination are addressed early. Operators and providers should therefore plan V2G and depot storage as an operational system, not as a technology add-on. This means agreeing in advance on dispatch rules (when charging and discharging is allowed), data-sharing arrangements, and how savings and flexibility revenues are shared between the operator, platform provider, and utility. Without these agreements, the business case is likely to remain limited to demonstrations.

• For industrial users, especially in mining and remote operations, the results indicate that SL-BESS can support stable operations and lower-carbon energy supply when offered as a service with clear lifecycle responsibility. A practical recommendation is to start with well-defined operational requirements—power profiles, uptime targets, mobility needs, and reporting requirements—and to use these as the basis for service-level agreements. This reduces uncertainty for both provider and customer and supports stepwise scaling as battery supply and standards mature.

Across all contexts, a key managerial implication is that SL-BESS should be integrated into planning for charging infrastructure and operations, not treated as a separate sustainability initiative. When deployed with clear ownership, monitoring, and revenue logic, second-life storage can reduce peak demand, buffer energy variability, and support more resilient and flexible operations. This strengthens the case for viewing circular energy storage as part of future production and infrastructure planning, particularly as electrification and regulatory requirements continue to intensify.

5.3 Limitations and future work

Several limitations should be acknowledged. First, the empirical scope is limited to Swedish stakeholders, which may reduce direct generalization to other contexts. However, because battery governance and sustainability requirements are increasingly shaped by European Union regulation, the insights are likely to be relevant beyond Sweden. Second, the scenarios are conceptual and grounded in qualitative evidence; this study does not include operational pilots,

detailed economic calculations, or quantified lifecycle performance assessments. Third, most participating organizations can be considered early adopters, which may bias the findings toward more positive expectations of second-life solutions.

In addition, the regulatory framework for batteries is evolving. The European Union's Battery Regulation introduces strengthened requirements related to traceability, sustainability reporting, and extended producer responsibility, which may affect mandates, ownership models, and the availability of batteries for second-life applications. As these requirements are still being implemented, their practical implications for second-life business models remain uncertain and were not fully captured in this study.

Future research should build on these findings in three directions. Quantitative work is needed to test the economic feasibility of SL-BESS business models under different electricity prices, degradation rates, and regulatory conditions, including sensitivity analyses. Environmental assessment could be strengthened by combining business modeling with life cycle assessment data to better quantify climate and resource benefits across scenarios. Finally, as second-life batteries become more tradable, research should examine market formation in greater detail, including standards and certification, data sharing and governance (such as traceability requirements and producer responsibility), and user acceptance among municipalities, fleet operators, and industrial customers.

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